

Phase 1 – Brainstorming and Ideation

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Problem Statement

In a system with multiple users and records, controlling access only through user roles may not always be sufficient. In some scenarios, access to a record needs to be controlled dynamically based on the value of a specific field within that record.

For example, records with different states may require different access permissions. A record with a particular field value may need to be accessible, while another record with a restricted value should not be accessible.

Therefore, a mechanism is required to evaluate a field value and dynamically allow or restrict access to the record.

2. Project Idea

The proposed project uses a **Script-Controlled Access Control List (ACL)** to control record access based on a specific field value.

The ACL will execute a script whenever a user attempts to access a record. The script checks the value of the specified field and determines whether the user should be allowed to access the record.

Basic Flow

User Requests Record

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ACL is Triggered

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Script Checks Field Value

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Access Decision is Made

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Allow Access / Restrict Access

3. Objective

The main objective of this project is to implement a script-controlled ACL that dynamically restricts record access based on the value of a specific field.

The project aims to:

- Control record-level access dynamically.
- Evaluate a field value before granting access.
- Allow access when the field value satisfies the defined condition.
- Restrict access when the field value does not satisfy the condition.
- Demonstrate how scripting can be used to implement dynamic access control.

4. Example Scenario

Consider a record containing a **State** field.

State	Access
Open	Allowed
In Progress	Allowed
Closed	Restricted

When a user attempts to access a record, the ACL script checks the value of the **State** field.

- If the state is **Open**, access is allowed.
- If the state is **In Progress**, access is allowed.
- If the state is **Closed**, access is restricted.

This demonstrates how record access can be controlled based on the value stored within the record.

5. Expected Outcome

At the end of this project, the system should be able to evaluate a record's field value through an ACL script and make an access decision accordingly.

The expected behavior is:

Valid Field Value → Access Granted

Restricted Field Value → Access Denied

6. Conclusion

During the brainstorming and ideation phase, the problem of dynamically controlling record access based on field values was identified. The proposed solution is to use a script-controlled ACL that evaluates the required field and determines whether access should be granted or restricted.

This phase establishes the foundation for the subsequent phases of the project, including requirement analysis, design, implementation, testing, and deployment.